

Amortized Simulation-Based Inference for Oncology PKPD

A Grounded Methods-Transfer Project Plan:
From Quantum Device Inference to Tumor-Growth Model Calibration

Simeoni Tumor Growth Models, Neural Posterior Estimation, Hierarchical Partial Pooling, and
Optional Gradient-Informed Extensions

Edoardo Spigarolo
EPFL – LTPN / AI Center
edoardo.spigarolo@gmail.com

May 2026 — Living Document

Abstract

This document is a grounded project plan for applying amortized simulation-based inference (SBI) to oncology pharmacokinetic–pharmacodynamic (PKPD) tumor-growth models. The project transfers methods from quantum device characterization—namely simulator-based parameter inference, posterior approximation, differentiable ODE solving, and hierarchical uncertainty modeling—to a pharma-relevant problem: estimating patient-level and population-level tumor response parameters from sparse longitudinal tumor-volume measurements. Unlike broader ML-for-drug-discovery plans based on protein–ligand GNN scoring, RL compound screening, or differentiable binding free energy, this plan is intentionally narrow. The core deliverable is a compact, publishable, end-to-end pipeline built around one validated simulator class, one inference family, and one evaluation protocol. Speculative extensions such as GNN molecular scoring, RL-guided screening, and differentiable molecular simulation are retained only as post-project research directions.

Contents

1	Strategic Framing	3
1.1	The Revised Core Thesis	3
1.2	Why the Previous GNN–RL–Differentiable-FEP Plan Was Too Broad	3
1.3	The Main Deliverable	4
1.4	What This Project Is Not	4
2	Core Problem: Oncology PKPD Tumor-Growth Inference	4
2.1	The Pharmacometric Setting	4
2.2	Simeoni-Style Tumor-Growth Model	5
2.3	Hierarchical Population Model	5
3	Inference Method: Neural Posterior Estimation	6
3.1	Why SBI Fits This Problem	6
3.2	Amortized Neural Posterior Estimation	6
3.3	Sequential NPE for Sharper Posteriors	6
3.4	Hierarchical NPE	7

4	Data Strategy	7
4.1	Week-1 Data Verification Gate	7
4.2	Fallback Strategy: Synthetic-First Validation	7
4.3	Target Dataset Hierarchy	8
5	Implementation Plan	8
5.1	Repository Structure	8
5.2	Minimal Simulator Scaffold	9
5.3	Dosing Discontinuity Handling	10
5.4	NPE Training Scaffold	10
5.5	Trajectory Encoding	11
6	Validation Protocol	11
6.1	Simulation-Based Calibration	11
6.2	Posterior Predictive Checks	11
6.3	Baseline Comparisons	11
6.4	Identifiability Diagnostics	12
7	Project Phases and Timeline	12
7.1	Phase 0: Data and Scope Gate	12
7.2	Phase 1: Synthetic Simulator and Individual NPE	13
7.3	Phase 2: Hierarchical NPE and Partial Pooling	13
7.4	Phase 3: Optional Gradient-Informed SBI	13
7.5	Phase 4: Manuscript and Release	14
8	Publication Strategy	14
8.1	Primary Paper	14
8.2	Possible Venues	14
8.3	Coauthor Strategy	15
9	Risk Assessment	15
10	Self-Teaching Curriculum	15
10.1	Priority Reading List	15
10.2	Software Proficiency Milestones	16
11	Future Research Directions After the Grounded Project	16
11.1	Direction F1: Molecular GNNs as Covariates for PKPD Response	16
11.2	Direction F2: Active Learning for Adaptive Virtual Trial Design	17
11.3	Direction F3: Gradient-Informed SBI and Sensitivity-Aware Experimental Design	17
11.4	Direction F4: Mechanistic–Molecular Bridge	17
11.5	Direction F5: Return to GNN/RL/FEP Only After the SBI Core Is Stable	17
12	Concrete Success Criteria	18
13	Key References	18

1 Strategic Framing

1.1 The Revised Core Thesis

The central methodological transfer is not “quantum computing for drug discovery.” It is **inference through simulators**. In quantum device characterization, the object of interest is a set of hidden physical parameters—noise rates, coupling strengths, loss channels, or calibration drifts—that are inferred from indirect measurements generated by a simulator or physical device. In oncology PKPD, the object of interest is also a set of hidden physical and biological parameters—tumor growth rate, drug kill rate, inter-patient variability, residual noise, and treatment-response delay—inferred from sparse longitudinal measurements.

The project therefore transfers the following working pattern:

Quantum Device Inference	→	Oncology PKPD Inference
Hamiltonian / Lindblad simulator		Tumor-growth ODE simulator
Hidden noise or coupling parameters		Hidden growth, kill, delay, and variability parameters
Measurement outcomes over time		Tumor-volume measurements over time
Posterior over physical parameters		Posterior over patient and population PKPD parameters
Amortized inference for repeated calibration		Amortized inference for repeated patient-level estimation

Key Insight

The grounded project is **not** a general ML drug-discovery pipeline. It is a focused SBI pipeline for oncology tumor-growth models. This makes the scope realistic: one simulator, one data type, one inference problem, one validation stack, and one publishable methodological contribution.

1.2 Why the Previous GNN–RL–Differentiable-FEP Plan Was Too Broad

The earlier plan attempted to connect three separate components:

1. GNN-based protein–ligand scoring,
2. RL-guided active compound screening,
3. differentiable molecular simulation for force-field or RBFE optimization.

Each component is valuable as a learning direction, but the integrated plan had three major assumption failures.

Honesty Check

Assumption failure 1: A single-person PaiNN/PDBbind scoring project is unlikely to produce a competitive scoring function in a few weeks. Protein–ligand binding-affinity prediction remains difficult even for specialized industrial and academic teams. A likely outcome is matching or underperforming established docking/scoring baselines on CASF-style benchmarks.

Assumption failure 2: Production-quality differentiable relative binding free energy (RBFE) is not feasible as a side-project deliverable. Real RBFE requires explicit-solvent molecular dynamics, validated force fields, alchemical transformations, sampling protocols, and careful uncertainty estimation. JAX-MD toy Lennard-Jones gradients do not directly translate to differentiable production FEP.

Assumption failure 3: RL compound screening is an interesting hypothesis, but it is not yet a working dot. Classical active learning and Bayesian optimization baselines are already strong. An RL policy may or may not improve over them, and proving that would be a separate research project.

The corrected strategy is to keep those ideas as future extensions, while moving the main project to a domain where the simulator, inference task, and validation protocol are all established.

1.3 The Main Deliverable

The main deliverable is an end-to-end open-source pipeline:

Dose schedule + observation times → Simeoni-style tumor-growth simulator → synthetic or real longitudinal tumor-volume data → neural posterior estimation → patient-level and population-level parameter posteriors → calibration, coverage, and NONMEM/Monolix-style comparison.

This is narrow enough to execute at 8–10 hours per week, but still scientifically meaningful. The result can be framed for a pharmacometrics audience as a comparison between amortized neural inference and standard nonlinear mixed-effects workflows.

1.4 What This Project Is Not

Honesty Check

This project does **not** claim that SBI replaces clinical pharmacometrics workflows immediately. It does not claim clinical utility from a toy synthetic benchmark. It does not claim patient-level precision when the data are sparse and non-identifiable. The aim is more modest and more defensible: demonstrate when amortized NPE reproduces calibrated posterior uncertainty for a standard oncology PKPD model, quantify when hierarchical partial pooling helps, and identify the regimes where neural inference fails or needs more careful modeling.

2 Core Problem: Oncology PKPD Tumor-Growth Inference

2.1 The Pharmacometric Setting

In oncology PKPD, one often observes tumor burden at a small number of time points. For patient i , the data may look like

$$\mathcal{D}_i = \{(t_{ij}, y_{ij})\}_{j=1}^{n_i}, \quad (1)$$

where t_{ij} is a scan or measurement time and y_{ij} is a tumor-volume or lesion-size measurement. The underlying biological trajectory is not directly observed. It is modeled through a mechanistic ODE parameterized by patient-specific parameters θ_i and population-level parameters ψ .

The statistical target is the posterior

$$p(\theta_i, \psi \mid \mathcal{D}_{1:N}, u_{1:N}), \quad (2)$$

where $u_i(t)$ denotes the treatment or dosing schedule for patient i .

2.2 Simeoni-Style Tumor-Growth Model

A standard model class for xenograft or oncology tumor-growth inhibition uses a proliferating tumor compartment and a chain of damaged or dying compartments. A representative form is

$$\frac{dx_1}{dt} = g(x_1; \lambda_0, \lambda_1, \psi_g) - k_{\text{kill}} C(t) x_1, \quad (3)$$

$$\frac{dx_2}{dt} = k_{\text{kill}} C(t) x_1 - k_{\text{tr}} x_2, \quad (4)$$

$$\frac{dx_3}{dt} = k_{\text{tr}} (x_2 - x_3), \quad (5)$$

$$\frac{dx_4}{dt} = k_{\text{tr}} (x_3 - x_4), \quad (6)$$

$$V(t) = x_1(t) + x_2(t) + x_3(t) + x_4(t), \quad (7)$$

where x_1 is the proliferating tumor compartment, x_2, x_3, x_4 are treatment-damaged compartments, $C(t)$ is the drug concentration or exposure proxy, k_{kill} controls drug effect, and k_{tr} controls the delay between drug-induced damage and observable tumor-volume reduction.

The growth function g can be chosen depending on the dataset:

$$\text{exponential growth: } g(x_1) = \lambda_0 x_1, \quad (8)$$

$$\text{Gompertz/logistic-like saturation: } g(x_1) = \frac{\lambda_0 x_1}{[1 + (\lambda_0 / \lambda_1 x_1)^{\psi_g}]^{1/\psi_g}}. \quad (9)$$

The observation model is commonly log-normal:

$$\log y_{ij} = \log V_i(t_{ij}; \theta_i, u_i) + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma_{\text{obs}}^2). \quad (10)$$

Key Insight

The simulator is simple enough to run thousands to millions of times, but expressive enough to capture the essential PKPD features needed for a publishable methodological benchmark: nonlinear growth, treatment effect, delayed response, sparse observations, inter-individual variability, and non-identifiability.

2.3 Hierarchical Population Model

For patient i , define patient-specific parameters on the log scale:

$$\log \theta_i \sim \mathcal{N}(\mu, \Sigma), \quad (11)$$

where μ and Σ are population-level parameters. The full model is

$$\mu, \Sigma, \sigma_{\text{obs}} \sim p(\mu, \Sigma, \sigma_{\text{obs}}), \quad (12)$$

$$\theta_i \mid \mu, \Sigma \sim \text{LogNormal}(\mu, \Sigma), \quad (13)$$

$$y_{ij} \mid \theta_i, u_i, t_{ij} \sim \text{LogNormal}(\log V_i(t_{ij}; \theta_i, u_i), \sigma_{\text{obs}}^2). \quad (14)$$

This is exactly the structure of nonlinear mixed-effects (NLME) pharmacometrics models. The SBI contribution is to amortize posterior inference for this model class.

3 Inference Method: Neural Posterior Estimation

3.1 Why SBI Fits This Problem

Classical Bayesian inference requires evaluating or approximating the likelihood

$$p(\mathcal{D} \mid \theta) = \prod_{i,j} p(y_{ij} \mid \theta_i, t_{ij}, u_i), \quad (15)$$

and then sampling the posterior using MCMC, Laplace approximation, or specialized NLME tools such as NONMEM, Monolix, or Stan.

Simulation-based inference instead uses the simulator directly:

$$\theta^{(m)} \sim p(\theta), \quad (16)$$

$$\mathcal{D}^{(m)} \sim p(\mathcal{D} \mid \theta^{(m)}), \quad (17)$$

$$q_\phi(\theta \mid \mathcal{D}) \approx p(\theta \mid \mathcal{D}). \quad (18)$$

The neural network q_ϕ is trained once on simulated pairs $(\theta^{(m)}, \mathcal{D}^{(m)})$ and then reused for fast posterior inference on new patients or new datasets.

3.2 Amortized Neural Posterior Estimation

The first core model is neural posterior estimation (NPE). It learns a conditional density estimator

$$q_\phi(\theta \mid x), \quad (19)$$

where x is an embedding of the observed tumor trajectory, dose schedule, and observation mask. The estimator can be a masked autoregressive flow, neural spline flow, normalizing flow, or flow-matching posterior network.

The training loss is

$$\mathcal{L}(\phi) = -\frac{1}{M} \sum_{m=1}^M \log q_\phi(\theta^{(m)} \mid x^{(m)}), \quad (20)$$

where each pair $(\theta^{(m)}, x^{(m)})$ is generated by the simulator.

3.3 Sequential NPE for Sharper Posteriors

If prior simulations are too diffuse, sequential NPE can focus simulations around high-posterior regions:

$$\theta^{(m)} \sim \tilde{p}_r(\theta), \quad x^{(m)} \sim p(x \mid \theta^{(m)}), \quad (21)$$

where $\tilde{p}_r$ is the proposal distribution at round r . Proposal correction is required so that the learned posterior remains calibrated under the original prior.

Honesty Check

Sequential NPE is a likely implementation requirement, not an optional luxury. PKPD posteriors can contain ridges, boundary effects, and strong correlations. A one-shot posterior trained from a broad prior may work on easy synthetic cases but fail on sparse or non-identifiable patient trajectories.

3.4 Hierarchical NPE

The second core contribution is hierarchical inference. The model should support both:

1. **individual NPE:** infer θ_i from patient i alone;
2. **hierarchical NPE:** infer θ_i while sharing information through population parameters (μ, Σ) .

The hypothesis is that hierarchical partial pooling improves inference when each patient has very few observations. This is especially relevant when patients have only 2–3 tumor-volume measurements.

Key Insight

The expected publishable result is not merely “NPE works.” The important question is: **When does hierarchical amortized inference improve patient-level parameter recovery under sparse longitudinal observations?**

4 Data Strategy

4.1 Week-1 Data Verification Gate

Before writing substantial model code, the first task is to verify whether usable longitudinal patient-level tumor-volume data are available.

Honesty Check

Open oncology PKPD data availability is the main early project risk. Some influential PKPD papers report population-level fits and summary statistics but do not release individual patient trajectories. The project should not assume that Claret-style or Stein-style patient-level datasets are directly available without checking supplementary materials, repositories, and example datasets.

The Week-1 gate is:

Data source	What to verify	Decision
Published PKPD papers	Are individual tumor-time curves available?	Use if real trajectories are downloadable
Monolix / pharmacometrics examples	Are oncology tumor-growth datasets included?	Use as open benchmark if available
Digitized literature figures	Are curves clear enough to digitize?	Use only if metadata are sufficient
Fully synthetic data	Can the simulator generate realistic sparse trajectories?	Always available fallback

4.2 Fallback Strategy: Synthetic-First Validation

If patient-level real data are not available, the project remains viable as a synthetic-but-realistic validation study. This is acceptable if the validation is rigorous:

- draw population parameters from pharmacologically plausible priors;
- simulate virtual trials with realistic observation schedules;
- include missingness, measurement noise, dropout, and sparse sampling;
- test posterior calibration using simulation-based calibration;
- compare NPE against MCMC or a NONMEM/Monolix-style baseline on the same synthetic data;
- clearly state that the result is a methodological benchmark, not a clinical model.

4.3 Target Dataset Hierarchy

Dataset type	Use	Priority
Real patient-level tumor trajectories	External validation and pharmacometric credibility	Highest
Open pharmacometrics example datasets	Reproducible benchmark and tool comparison	High
Digitized curves from publications	Secondary validation when raw data are unavailable	Medium
Synthetic virtual patients	Controlled validation, SBC, ablations, identifiability studies	Mandatory

5 Implementation Plan

5.1 Repository Structure

Listing 1: Recommended repository layout

```

oncology-sbi-pkpd/
|-- README.md
|-- pyproject.toml
|-- environment.yml
|-- src/
|   |-- pkpd/
|   |   |-- simeoni.py           # Tumor-growth ODE simulator
|   |   |-- dosing.py           # Bolus/infusion schedules, piecewise
|   |       integration
|   |   |-- observation.py      # Log-normal observation model, masks
|   |-- data/
|   |   |-- loaders.py          # Real/open dataset loaders
|   |   |-- digitized.py        # Optional WebPlotDigitizer imports
|   |   |-- synthetic.py        # Virtual patient/trial generator
|   |-- inference/
|   |   |-- npe.py              # Individual NPE training
|   |   |-- sequential_npe.py   # SNPE-C/proposal correction
|   |   |-- hierarchical_npe.py # Population/partial-pooling model
|   |   |-- summaries.py        # Trajectory encoders and masks
|   |-- baselines/
|   |   |-- mcmc_stan.py         # Optional Stan/PyMC baseline
|   |   |-- nonmem_export.py     # Dataset export for NONMEM/Monolix
|   comparison
|   |-- validation/
|   |   |-- sbc.py              # Simulation-based calibration
|   |   |-- coverage.py         # Posterior interval coverage
|   |   |-- posterior_predictive.py # PPC plots and metrics
|   |   |-- identifiability.py   # Ridge/correlation diagnostics
|   |-- experiments/
|   |   |-- train_npe.py
|   |   |-- train_hierarchical.py
|   |   |-- run_sbc.py
|   |   |-- compare_baselines.py
|-- configs/
|   |-- simulator_default.yaml
|   |-- npe_default.yaml
|   |-- hierarchical_default.yaml
|-- notebooks/
|-- tests/

```

```
|-- docs/
```

5.2 Minimal Simulator Scaffold

Listing 2: Simeoni-style ODE simulator scaffold using JAX/Diffrax-style structure

```
import jax
import jax.numpy as jnp

def growth_term(x1, lam0, lam1, psi_g=1.0):
    """
    Saturating growth term. For early tumor growth this behaves
    approximately exponentially; for large tumor burden it slows down.
    """
    denom = (1.0 + (lam0 / lam1 * jnp.maximum(x1, 1e-8)) ** psi_g) **
            (1.0 / psi_g)
    return lam0 * x1 / denom

def concentration_from_schedule(t, dose_times, dose_amounts, kel):
    """
    Simple one-compartment exposure proxy.
    Production version should support piecewise integration across dose
    events.
    """
    dt = t - dose_times
    active = dt >= 0.0
    return jnp.sum(jnp.where(active, dose_amounts * jnp.exp(-kel * dt),
                             0.0))

def simeoni_rhs(t, x, params, schedule):
    """
    x = [x1, x2, x3, x4]
    params include lam0, lam1, psi_g, k_kill, k_tr, kel.
    schedule includes dose_times and dose_amounts.
    """
    x1, x2, x3, x4 = x
    C_t = concentration_from_schedule(
        t,
        schedule["dose_times"],
        schedule["dose_amounts"],
        params["kel"],
    )

    grow = growth_term(x1, params["lam0"], params["lam1"], params["psi_g"]
                       ])
    kill = params["k_kill"] * C_t * x1
    ktr = params["k_tr"]

    dx1 = grow - kill
    dx2 = kill - ktr * x2
    dx3 = ktr * (x2 - x3)
    dx4 = ktr * (x3 - x4)
    return jnp.array([dx1, dx2, dx3, dx4])
```

```
def observe_tumor_volume(states, sigma_obs, key):
    """Log-normal observation model."""
    true_volume = jnp.sum(states, axis=-1)
    eps = sigma_obs * jax.random.normal(key, shape=true_volume.shape)
    return jnp.exp(jnp.log(jnp.maximum(true_volume, 1e-8)) + eps)
```

5.3 Dosing Discontinuity Handling

Drug dosing creates discontinuities. Bolus doses are not smooth functions of time, and infusion schedules can introduce step changes. The clean implementation is piecewise integration:

1. sort all dose and observation times;
2. integrate the ODE between event times;
3. apply dose updates exactly at dose events;
4. record states at observation times;
5. concatenate the full trajectory.

Honesty Check

Do not hide dosing discontinuities inside a naive smooth function unless the approximation is explicitly part of the experiment. Incorrect handling of dose events can create biased gradients, unstable training data, or misleading posterior fits.

5.4 NPE Training Scaffold

Listing 3: High-level NPE training skeleton

```
from sbi.inference import SNPE
from sbi.utils import BoxUniform
import torch

def simulator(theta):
    """
    Convert parameter vector theta into a simulated observation vector.
    Includes:
    - patient-specific PKPD parameters
    - dose schedule
    - observation times
    - log-normal observation noise
    - missingness mask if required
    """
    x = simulate_patient_trajectory(theta)
    return torch.as_tensor(x, dtype=torch.float32)

# Example prior on transformed/log parameters.
prior = BoxUniform(
    low=torch.tensor([-6.0, -6.0, -8.0, -6.0, -6.0, -4.0]),
    high=torch.tensor([ 2.0,  4.0,  1.0,  2.0,  2.0,  1.0]),
)

inference = SNPE(prior=prior)

theta = prior.sample((50_000,))
x = torch.stack([simulator(th) for th in theta])

inference = inference.append_simulations(theta, x)
```

```
density_estimator = inference.train()
posterior = inference.build_posterior(density_estimator)

# Later: posterior.sample((10_000,), x=observed_patient_vector)
```

5.5 Trajectory Encoding

The observation vector x must encode irregular time series. Three encoders should be implemented in increasing sophistication:

Encoder	Description	Use
Fixed-grid interpolation	Interpolate tumor volume onto a common time grid and include an observation mask	Simple baseline
Set encoder	Encode unordered (t, y, mask) pairs with DeepSets or attention pooling	Robust to irregular observations
Neural controlled differential equation	Continuous-time sequence encoder	Optional advanced extension

The first paper should not depend on the most advanced encoder. A fixed-grid plus mask baseline is enough for Phase 1; a set encoder is enough for Phase 2.

6 Validation Protocol

6.1 Simulation-Based Calibration

Simulation-based calibration (SBC) is mandatory. The validation loop is:

1. sample $\theta^{(m)} \sim p(\theta)$;
2. simulate $x^{(m)} \sim p(x \mid \theta^{(m)})$;
3. sample $\tilde{\theta}^{(m,1:K)} \sim q_\phi(\theta \mid x^{(m)})$;
4. compute the rank of the true parameter among posterior samples;
5. check whether ranks are uniform.

If ranks are not uniform, the posterior is miscalibrated. The failure mode should be reported, not hidden.

6.2 Posterior Predictive Checks

For each observed patient trajectory, sample parameters from the posterior and simulate replicated trajectories:

$$\theta^{(k)} \sim q_\phi(\theta \mid x_{\text{obs}}), \quad \tilde{x}^{(k)} \sim p(x \mid \theta^{(k)}). \quad (22)$$

The posterior predictive distribution should be compared against the observed tumor curve. Metrics:

- coverage of observed points by posterior predictive intervals;
- RMSE/MAE of posterior predictive median;
- calibration of 50%, 80%, and 95% intervals;
- failure cases under sparse observation schedules.

6.3 Baseline Comparisons

The baseline comparison should be staged:

Baseline		Purpose	Priority
MCMC	in	Gold-standard posterior on small synthetic cases	Mandatory
Stan/PyMC			
Laplace / MAP fit		Fast classical approximation	Useful
NONMEM	or	Pharmacometric credibility	Strongly preferred
Monolix			
Individual-only NPE		Ablation for hierarchical model	Mandatory

Key Insight

For a pharmacometrics audience, the key comparison is not against a generic neural network. The key comparison is against the familiar workflow: nonlinear mixed-effects estimation, posterior uncertainty, shrinkage/partial pooling, and predictive checks.

6.4 Identifiability Diagnostics

PKPD models often exhibit practical non-identifiability. Different parameter combinations can generate nearly identical tumor curves. The project should explicitly diagnose:

- posterior correlations between growth and kill parameters;
- weak identifiability of delay-chain parameters under sparse observations;
- boundary effects for parameters constrained to be positive;
- multimodality under ambiguous response patterns;
- sensitivity to number and timing of tumor measurements.

A negative result in this section is valuable. If a parameter is not identifiable from 2–3 measurements, the correct output is a broad posterior, not a falsely precise point estimate.

7 Project Phases and Timeline

7.1 Phase 0: Data and Scope Gate

Week	Task	Details	Deliverable
1	Data verification	Check raw/supplementary oncology PKPD datasets, Monolix examples, digitizable curves, and synthetic fallback assumptions	Data decision memo
1	Literature framing	Read Simeoni-style tumor models, NLME basics, SBI/NPE papers, and pharmacometrics validation norms	Annotated bibliography

7.2 Phase 1: Synthetic Simulator and Individual NPE

Week	Task	Details	Deliverable
2	Simulator	Implement ODE, dosing schedules, observation model, priors, virtual patients	Tested simulator
3	Data generator	Generate virtual trials with sparse observation schedules and known parameters	Synthetic dataset suite
4	Individual NPE	Train one-shot NPE on patient-level parameters	First posterior model
5	Calibration	Run SBC and posterior predictive checks	Calibration report
6	Baseline MCMC	Compare against Stan/PyMC on small cases	Baseline comparison

7.3 Phase 2: Hierarchical NPE and Partial Pooling

Week	Task	Details	Deliverable
7	Population model	Add population-level priors and inter-individual variability	Hierarchical simulator
8	Hierarchical encoder	Encode multi-patient datasets and patient masks	Dataset encoder
9	Hierarchical NPE	Train partial-pooling posterior estimator	Hierarchical model
10	Sparse-data ablation	Compare 2, 3, 4, 6+ observations per patient	Partial-pooling curves
11	External/open data	Fit available real or example dataset, if verified in Week 1	Real-data case study
12	Paper figure freeze	Freeze main figures: SBC, coverage, PPC, ablations, baseline comparison	Figure set

7.4 Phase 3: Optional Gradient-Informed SBI

Gradient-informed SBI should be treated as a bonus contribution, not a core dependency. The hypothesis is that gradients through the ODE simulator can reduce simulation budget or improve posterior learning. This may work, but the gain is uncertain for PKPD posteriors with ridges, multimodality, and stiff or discontinuous dynamics.

Week	Task	Details	Deliverable
13	Differentiable solver	Implement or verify differentiable ODE path with event-aware dosing	Gradient test
14	Gradient features	Add score/Jacobian/sensitivity features to NPE training	Experimental model
15	Budget ablation	Compare simulation budgets: 5k, 10k, 25k, 50k, 100k	Efficiency curves
16	Decide inclusion	Include only if calibration improves or simulation budget drops meaningfully	Go/no-go decision

Honesty Check

The project must remain publishable if Phase 3 fails. The core paper is Phase 1 + Phase 2: calibrated individual and hierarchical amortized inference for a standard oncology PKPD model.

7.5 Phase 4: Manuscript and Release

Week	Task	Details	Deliverable
17–18	Manuscript draft	Write methods, validation, results, limitations, and pharmacometric framing	Full draft
19	Code release	Clean configs, tests, notebooks, and reproduction scripts	GitHub release
20	Submission prep	Select venue and adapt framing	Submission package

8 Publication Strategy**8.1 Primary Paper**

Working title: *Amortized Hierarchical Inference for Oncology Tumor-Growth Models Under Sparse Longitudinal Observations*

Core claim: Neural posterior estimation can produce calibrated patient-level and population-level posterior uncertainty for Simeoni-style oncology PKPD models, with hierarchical partial pooling improving inference when individual patients have sparse tumor-volume measurements.

Main figures:

1. schematic of simulator and amortized inference pipeline;
2. posterior recovery on synthetic patients;
3. SBC rank histograms for individual NPE;
4. comparison against MCMC/NLME baseline;
5. hierarchical vs individual NPE under sparse observations;
6. posterior predictive checks on real or example data;
7. optional simulation-budget reduction from gradient-informed features.

8.2 Possible Venues

Venue	Fit	Framing
CPT: Pharmacometrics & Systems Pharmacology	Strong pharmacometrics fit, conservative audience	NLME comparison and clinical-modeling language
Journal of Pharmacokinetics and Pharmacodynamics	Direct PKPD methodology fit	Population modeling and uncertainty calibration
NeurIPS / ICML AI4Science workshop	Strong ML-methods fit	Amortized inference and simulator-based learning
JOSS	Software paper after method release	Reproducible open-source implementation

Key Insight

For pharmacometrics venues, avoid over-centering terms like “normalizing flows” or “AI pipeline” in the title. Lead with familiar concepts: oncology tumor-growth models, sparse longitudinal data, hierarchical inference, nonlinear mixed-effects comparison, and posterior predictive validation.

8.3 Coauthor Strategy

The project will be stronger with a pharmacometrics or oncology modeling coauthor. A useful collaborator can help with:

- selecting realistic priors and dosing schedules;
- interpreting NONMEM/Monolix baselines;
- checking whether the validation is credible to pharma reviewers;
- identifying real or semi-real open datasets;
- preventing language that sounds impressive to ML readers but naive to pharmacometricians.

A BMS collaborator does not need to share proprietary data for the first paper. Target selection, domain feedback, and validation of assumptions would already be valuable.

9 Risk Assessment

Risk		Prob.	Impact	Mitigation
No open patient-level data		Medium	Medium	Use synthetic-first validation; add Monolix/example data; digitize only if metadata are adequate
NPE is miscalibrated		Medium	High	Use SBC early; move to sequential NPE; narrow priors; improve trajectory encoder
Parameters are non-identifiable		High	Medium	Report broad posteriors; analyze identifiability; emphasize uncertainty rather than point recovery
Hierarchical gain is small		Medium	Medium	Report regimes where individual NPE is sufficient; focus on sparse-data thresholds
Gradient-informed SBI does not help		Medium	Low	Keep it optional; paper stands on Phases 1–2
Pharmacometrics reviewers are skeptical	re-	Medium	Medium	Include NONMEM/Monolix-style baseline and domain coauthor if possible
8–10 hrs/week is exceeded	ex-	Medium	Medium	Keep one simulator and one inference stack; postpone molecular GNN/RL/FEP ideas

10 Self-Teaching Curriculum**10.1 Priority Reading List**

1. **Tumor-growth inhibition models:** Simeoni et al., “Predictive pharmacokinetic-pharmacodynamic modeling of tumor growth kinetics in xenograft models after administration of anticancer agents,” 2004.

2. **Population PKPD / NLME basics:** Bonate, *Pharmacokinetic-Pharmacodynamic Modeling and Simulation*; or Gabrielsson and Weiner, *Pharmacokinetic and Pharmacodynamic Data Analysis*.
3. **Model evaluation:** papers/tutorials on visual predictive checks, posterior predictive checks, shrinkage, and parameter identifiability in NLME models.
4. **SBI foundations:** Cranmer, Brehmer, and Louppe, “The frontier of simulation-based inference,” 2020.
5. **NPE/SNPE:** Papamakarios and Murray, sequential neural posterior estimation; Greenberg et al., automatic posterior transformation for likelihood-free inference.
6. **SBI software:** the `sbi` package tutorials, especially examples involving ODE or epidemiological simulators.
7. **Hierarchical SBI / pharmacology:** recent work on amortized or hierarchical SBI for pharmacological nonlinear mixed-effects models.
8. **Differentiable ODE solving:** Diffrax documentation and adjoint-method references for event-aware ODE integration.

10.2 Software Proficiency Milestones

Tool	Milestone	Est. Time
JAX / NumPyro basics	Vectorize parameter sampling and simulator calls	1–2 days
Diffrax or SciPy ODE	Solve tumor-growth ODE and handle dosing events	2–3 days
<code>sbi</code>	Train NPE on a toy ODE simulator and run posterior sampling	2 days
Stan or PyMC	Fit a small synthetic patient dataset with MCMC	2–3 days
ArviZ / diagnostics	Plot posterior intervals, rank histograms, trace plots, coverage	1 day
Hydra / YAML configs	Reproducible experiment sweeps	1 day

11 Future Research Directions After the Grounded Project

This section preserves the strongest ideas from the earlier ML drug-discovery plan, but correctly labels them as **post-project research directions**. They should not be required for the first paper.

11.1 Direction F1: Molecular GNNs as Covariates for PKPD Response

Future Research Direction

Instead of training a GNN to solve general protein–ligand binding affinity from PDBbind, use molecular representations as **covariates inside the PKPD/SBI pipeline**. For example, if multiple compounds are available, encode each compound using RDKit fingerprints, ChemBERTa embeddings, or a molecular GNN, and let the drug-effect parameter depend on compound features:

$$\log k_{\text{kill}} = h_{\eta}(z_{\text{mol}}) + \eta_i, \quad (23)$$

where z_{mol} is a molecular embedding and η_i is patient-level random effect.

This is much more grounded than claiming to beat docking scoring functions. The GNN is not asked to solve binding affinity from scratch; it is used to explain variation in PKPD response across compounds.

11.2 Direction F2: Active Learning for Adaptive Virtual Trial Design

Future Research Direction

The earlier RL screening idea can be reframed as **adaptive experimental design for tumor-response inference**. Instead of choosing which compound to synthesize or screen, the policy chooses which observation time, dose level, or virtual experiment to run next in order to reduce posterior uncertainty.

This preserves the useful transfer from adaptive quantum measurement scheduling, but attaches it to the grounded simulator. The action space becomes:

$$\mathcal{A} = \{\text{next observation time, dose schedule, patient subgroup, experimental condition}\}, \quad (24)$$

and the reward can be expected information gain or reduction in posterior entropy.

This is a stronger research direction than RL compound screening because the simulator is under your control and the information-theoretic objective is well-defined.

11.3 Direction F3: Gradient-Informed SBI and Sensitivity-Aware Experimental Design

Future Research Direction

If differentiable ODE inference works well, use sensitivities

$$S_{jk} = \frac{\partial V(t_j; \theta)}{\partial \theta_k} \quad (25)$$

for experimental design, posterior diagnostics, or training acceleration.

This could connect directly to Fisher information:

$$\mathcal{I}(\theta) = S(\theta)^\top \Sigma_{\text{obs}}^{-1} S(\theta), \quad (26)$$

which gives a classical criterion for choosing informative observation times. This is a clean bridge between differentiable simulation and adaptive scheduling without needing molecular dynamics.

11.4 Direction F4: Mechanistic–Molecular Bridge

Future Research Direction

Once the PKPD SBI pipeline is mature, add molecular or omics covariates to explain inter-patient or inter-compound variability. Examples include mutation status, pathway activation, baseline tumor burden, drug class, or molecular embeddings.

A possible hierarchical model is

$$\log \theta_i \sim \mathcal{N}(\mu_i, \Sigma), \quad (27)$$

$$\mu_i = \mu_0 + Bz_i, \quad (28)$$

where z_i contains patient, tumor, genomic, or compound covariates. This is a natural path toward translational oncology without overclaiming immediate drug-discovery impact.

11.5 Direction F5: Return to GNN/RL/FEP Only After the SBI Core Is Stable

The original ML drug-discovery components can still be studied, but with realistic positioning:

Original idea	Reframed as	When to attempt
PaiNN on PDBbind	Learning exercise or covariate model, not guaranteed scoring breakthrough	After SBI paper draft
RL compound screening	Active design over simulator or virtual-trial protocol	After validated posterior model
Differentiable RBFE	Long-term research, not side-project deliverable	Only with dedicated comp-chem collaboration

12 Concrete Success Criteria

The project is successful if, by the end of Phase 2, it produces:

1. a tested Simeoni-style simulator with realistic dosing and observation schedules;
2. an individual NPE model with calibrated posterior uncertainty on synthetic data;
3. a hierarchical NPE model showing where partial pooling helps or does not help;
4. SBC, posterior predictive checks, and identifiability diagnostics;
5. comparison against MCMC and, if possible, a NONMEM/Monolix-style workflow;
6. a clean repository with reproducible configs and scripts;
7. a manuscript draft framed in pharmacometric language.

The project is **not** judged by whether every parameter is precisely recovered. In sparse PKPD data, correct uncertainty is often more important than point accuracy. A calibrated broad posterior is a success; an overconfident wrong estimate is a failure.

13 Key References

- [1] M. Simeoni, P. Magni, C. Cammia, G. De Nicolao, V. Croci, E. Pesenti, M. Germani, I. Poggesi, and M. Rocchetti, “Predictive pharmacokinetic-pharmacodynamic modeling of tumor growth kinetics in xenograft models after administration of anticancer agents,” *Cancer Research*, 64(3), 1094–1101, 2004.
- [2] P. Bonate, *Pharmacokinetic-Pharmacodynamic Modeling and Simulation*, Springer, 2011.
- [3] J. Gabrielsson and D. Weiner, *Pharmacokinetic and Pharmacodynamic Data Analysis: Concepts and Applications*, Swedish Pharmaceutical Press, 2016.
- [4] K. Cranmer, J. Brehmer, and G. Louppe, “The frontier of simulation-based inference,” *Proceedings of the National Academy of Sciences*, 117(48), 30055–30062, 2020.
- [5] G. Papamakarios and I. Murray, “Fast epsilon-free inference of simulation models with Bayesian conditional density estimation,” *Advances in Neural Information Processing Systems*, 2016.
- [6] D. Greenberg, M. Nonnenmacher, and J. Macke, “Automatic posterior transformation for likelihood-free inference,” *International Conference on Machine Learning*, 2019.
- [7] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman, “Validating Bayesian inference algorithms with simulation-based calibration,” *arXiv:1804.06788*, 2018.
- [8] P. Kidger, “On neural differential equations,” PhD thesis, University of Oxford, 2021. See also the DiffraX software documentation.
- [9] R. Arruda et al., “Amortized Bayesian inference for pharmacological nonlinear mixed-effects models,” 2024.
- [10] J. Zeghal et al., “Gradient-informed simulation-based inference,” 2023.

- [11] M. Volkov, J. A. Turk, N. Drizard, et al., “On the Frustration to Predict Binding Affinities from Protein-Ligand Structures,” *Journal of Medicinal Chemistry*, 65(11), 7946–7958, 2022.
- [12] D. E. Graff, E. I. Shakhnovich, and C. W. Coley, “Accelerating high-throughput virtual screening through molecular pool-based active learning,” *Chemical Science*, 12, 7866–7881, 2021.